

Jeongwoo Choi

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Education

Yonsei University

Ph.D. Student in the School of Electrical and Electronic Engineering
Computer Vision Lab (Advisor: Prof. Bumsub Ham)

Seoul, Korea
Mar 2024 – Present

Yonsei University

B.S. in the School of Electrical and Electronic Engineering
Relevant Coursework: Application Programming, Introduction Artificial Intelligence, Deep Learning Lab

Seoul, Korea
Mar 2020 – Feb 2024

Research Experience

Image diffusion model quantization (with Samsung Electronics)

Mar 2024 – May 2025

- Developed novel PTQ techniques to specifically address the **accumulated error problem** in diffusion models.
- Designed and implemented an **end-to-end PTQ pipeline** during a research collaboration with Samsung Electronics.

DITTO: Doodle to Image TranslaTiOn (Personal project)

May 2023 – Jun 2023

- Developed a ControlNet-based **web application** for real-time sketch-to-image translation.
- Enhanced prompt alignment** and visual fidelity by fine-tuning ControlNet on the SBU Caption dataset.

Music generation from incomplete midi sequence (with POZALabs)

Jan 2023 – May 2023

- Developed deep learning models to **reconstruct complete melodies** from incomplete MIDI sequences.
- Proposed the **Musical Similarity Index Measure (MSIM)**, a novel metric for evaluating melody reconstruction quality.

Publications

Published / Accepted

Relational Feature Caching for Accelerating Diffusion Transformers (*ICLR 2026*)

- Proposed **Relational Feature Estimation (RFE)** leveraging input-output relations to reduce redundant computation.
- Proposed **Relational Cache Scheduling (RCS)** for adaptive computation based on estimated prediction errors.
- Achieved SOTA on DiT-XL/2, FLUX.1 dev, and Hunyuan Video while maintaining efficiency.

AccuQuant: Simulating Multiple Denoising Steps for Quantizing Diffusion Models (*NeurIPS 2025*)

- Proposed a PTQ method that **simulates the multi-step denoising process** to minimize accumulated quantization errors.
- Proposed a novel **gradient approximation technique**, reducing calibration memory complexity from $O(n)$ to $O(1)$.
- Achieved SOTA low-bit PTQ results on DDIM, Latent Diffusion Model, and Stable Diffusion, outperforming prior work.

Under review

Q-SAM: Quantizing Segment Anything Models

- Proposed **Outlier-Agnostic Quantization (OaQ)** and **Quantization-Aware Re-distribution (QaR)** to handle activation outliers and distribution gaps in the mask decoder of SAM.
- Introduced **Noise-Aware Prompt Refinement (NPR)** to filter inaccurate calibration prompts.
- Achieved SOTA on 4-bit SAM quantization, more than doubling mAP over prior method (e.g., 30.2 vs 13.4).

Teaching Experience

Data structure and Algorithms

Sep 2025 – Dec 2025

Deep Learning Lab

Mar 2025 – Jun 2025

Introduction Artificial Intelligence

Sep 2024 – Dec 2024

Skills & Interests

Technical: Python, PyTorch, CUDA, Git, Docker, Conda

Languages: Korean (Native), English (Fluent)

Research Interests: Generative models, Efficient AI (e.g., Quantization, Caching)